Tianyang Zhou

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Education

- **Ph.D. in Computer Science** Aug 2023 – May 2028 *(Expected)*
  University of Illinois Urbana-Champaign, Illinois, USA
  - ◇ Research Area: Program Translation, Software Security
  - ◇ Advisors: Varun Chandrasekaran, Kirill Levchenko
- **M.S. in Computer Engineering** Sep 2021 – Jun 2023
  University of California, San Diego, California, USA
  - ◇ Advisor: Deian Stefan
- **B.E. in Software Engineering** Sep 2017 – Jun 2021
  Xi'an Jiaotong University, Shaanxi, China

Professional Experience

- **SWE Intern @ Google SRG (SystemsResearch@Google)** May 2026 - Aug 2026
  - ○ Owned an end-to-end, spec-driven LLM agent pipeline that generates and validates VMM device code on Cloud Hypervisor.
  - ○ Generated 1.5k LoC virtio-nvme core and 1.2k LoC tests from ∼500 LoC spec at ∼80% coverage; merged into the dev branch.

Selected Publications

- **Verifying Security Preservation in Code Translation** *(In Progress)*
  Tianyang Zhou, Varun Chandrasekaran.
  - ○ Designing LLM-assisted specification extraction from C, with CBMC proofs on C and Kani checks on Rust rewrites.
  - ○ Piloting on GNU coreutils and its Rust rewrite uutils.
- **SoK: Understanding Security Measurement Gaps in AI4Code**
  Tianyang Zhou, Taoran Li, Qilong Wu, Varun Chandrasekaran.
  Under review at IEEE Conference on Secure and Trustworthy Machine Learning (SaTML) 2027.
  - ○ Led a review of 328 sampled studies and 70 benchmarks, identifying proxy, scope, and property gaps in security evaluation.
  - ○ Designed three empirical studies; an audit found vulnerabilities in 20 of 30 repository outputs that passed all benchmark checks.
- **Verifiers in the Loop: Decoding Time Verification for Code Translation**
  Tianyang Zhou, Somesh Jha, Mihai Christodorescu, Kirill Levchenko, Varun Chandrasekaran.
  Annual Conference on Neural Information Processing Systems (NeurIPS) 2026. arXiv:2605.17626
  - ○ Built a decoding-time verification framework that catches translation errors on-the-fly via compilation, lint, and test oracles.
  - ○ Demonstrated cross-language generalizability with end-to-end evaluation on C-to-Rust and JS-to-TS pipelines.
- **SACTOR: LLM-Driven Correct and Idiomatic C to Rust Translation with Static Analysis and FFI-Based Verification**
  Tianyang Zhou, Ziyi Zhang, Haowen Lin, Somesh Jha, Mihai Christodorescu, Kirill Levchenko, Varun Chandrasekaran.
  Association for Computational Linguistics (ACL) 2026. NeurIPS Workshop DL4C 2025. 2026.acl-long.28
  - ○ Designed and implemented a two-phase LLM with static analysis pipeline for C-to-Rust translation.
  - ○ Built FFI-based verification harness enabling scalable end-to-end equivalence testing across translation stages.
- **Fine-grained Library Sandboxing for Rust Ecosystem**
  Tianyang Zhou.
  Master's Thesis, University of California, San Diego. 2023. ProQuest

Technical Skills

- **Programming:** C, C++, Rust, Python
- **Security/PL Skills:** LLVM/Clang, Program Analysis, Fuzzing, Sanitizers
- **ML for Code:** Transformers, LLM Fine-tuning, Verifier-in-the-loop, Agentic Workflows

Invited Talks

- Toward secure, verifiable, and idiomatic LLM-assisted C-to-Rust translation — *Intel*, Oct 2026
- SACTOR: LLM-Driven Multi-Step Translation from C to Rust using Static Analysis — *Google Research*, May 2025

Academic Service

- **NeurIPS 2026**, *Reviewer*; recognized as a **Top Reviewer**